Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 18-Nov-2022

Time: Three Hours **Max. Marks: 75**

Medicinal Chemistry - I
Q.P. CODE: 5014

Your answers should be specific to the questions asked
Draw neat labeled diagrams wherever necessary
All the Questions are compulsory

LONG ESSAYS **2 x 10 = 20 Marks**

1. Define and classify beta adrenergic blocking agents and write their SAR. Outline the synthesis of propranolol.

   OR

   What are anticonvulsants? Classify chemically with an example each. Enumerate the structure, chemical name; synthesis and specific use any one anticonvulsant.

2. Define anti-inflammatory drugs. Write the structure and uses of any four such drugs. Write the synthesis of Ibuprofen.

SHORT ESSAYS **7 x 5 = 35 Marks**

3. Define and classify biotransformation of drugs. What is its importance? Write the sites of biotransformation.

   OR

   Write in detail ionization and solubility as an important physicochemical parameter in predicting in biological action of drugs.

4. Explain the importances of optical activity and bioisosterism in drug action.

   OR

   Explain the mechanism of action and uses a) Terbutaline b) Ephedrine c) Methysergide d) Atenolol

5. What are solanaceous alkaloids? Give examples. Write the synthesis and specific use of Ipratropium bromide.
6. Discuss the SAR of parasympathomimetic agents.
7. Discuss the role of acetylcholinesterase in the body. Classify acetylcholine inhibitors with two examples each along with their specific uses.
8. What are ultra-short acting barbiturates? How are they useful? Name any two of them with structures.
9. Discuss the SAR of Benzodiazepins. Write the structure and uses of Alprazolam.

SHORT ANSWERS **10 x 2 = 20 Marks**

10. Define Oxidation reactions in drug metabolism giving two examples
11. Write the diagrammatic representation of Cytochrome – P 450 role in drug metabolism
12. Write the structure and uses of Propronalal.
13. What is a catecholamine? Mention any two neurotransmitter catacholamines with their biological functions.
14. Write a note on alpha receptors.
15. Write a note on Nicotinic receptors.
16. Write the structure and uses of Tridihexethyl chloride.
17. Write the structure and uses of Isopropamide iodide.
18. Write the mechanism of action of Phenobarbitone and Alprazolam.
19. Write the structure and specific uses Loperamide hydrochloride.

*****